

Atul Krishnaa V

Software Engineer · Machine Learning · Full-Stack Development
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PROFESSIONAL SUMMARY

Results-driven Software Engineer and Machine Learning practitioner with hands-on experience delivering production systems at scale — including a customer data platform currently deployed in Nigeria, processing 10 million records per run. Interned at the National University of Singapore (Deep Learning, awarded High Commendation) and at a UK-backed AI start-up. Skilled across the full stack: FastAPI · Next.js 15 · React 19 · PyTorch · scikit-learn. Seeking graduate roles in software engineering, ML engineering, or full-stack product development.

EXPERIENCE

Indian Space Research Organisation (ISRO)

May 2026- Present

Big Data Analytics & Deep Learning Intern

- Architecting and training deep learning models for the semantic segmentation of Synthetic Aperture Radar (SAR) imagery.
- Developing robust data processing pipelines to handle complex, high-resolution geospatial satellite data.

National University of Singapore — Global Academic Internship Programme (GAIP)

Dec 2025

Big Data Analytics & Deep Learning Intern

- Developed and optimised predictive deep learning models using TensorFlow and PyTorch on large-scale datasets; applied feature engineering, hyperparameter tuning, and full evaluation pipelines (accuracy, F1, confusion matrix).
- Delivered technical findings to NUS faculty and industry reviewers; awarded High Commendation for project quality, rigour, and presentation clarity.

Dataspire Technologies

Jan 2025

Software Engineering Intern

- Engineered a production-ready, full-stack AI conversational web application using Azure OpenAI, owning backend API design, AI service integration, and a responsive React frontend from inception through deployment.

PROJECTS

Nexus 360 — Customer Data Activation & Behavioural Classification System

Apr 2026

Python 3.9 · Pandas · NumPy · Vectorised ML Pipeline · Client: MetaYB / Tolaram Group · Live in Nigeria

- Architected a scalable Customer 360 data pipeline processing 10 million records across 20 sequential campaign cycles, achieving 285,000 records/min throughput with a 35-minute end-to-end runtime at 4.2 GB peak memory.
- Designed a turnout-triggered enrichment engine that increased physical campaign conversion from 38.78% to 66.67% — a +27.89 percentage-point uplift (+71.9% relative improvement) and an average gain of +1.47 pp per cycle.
- Reduced memory consumption by ~60% through dtype engineering (int8 / int32 / float32 / category); delivered 50+ ML-ready feature columns, 20 binary turnout labels, and a 3.86× growth in profile density over baseline.

Public Access & Healthcare Readiness Platform

2025

Next.js 15 · React 19 · TypeScript · FastAPI · Redis · Open-Meteo · Overpass/Nominatim (OSM) · Tailwind CSS 4.0 · Vercel · Render

- Engineered a FastAPI decision engine correlating real-time weather data (Open-Meteo), OSM facility discovery (Overpass/Nominatim), and regional holiday calendars to predict overcrowding at public offices and hospitals across India.
- Reduced API response latency by 40% via Redis caching for location and weather data; implemented an Explainable AI (XAI) output layer surfacing transparent, factor-level reasoning for every recommendation.
- Architected a Next.js 15 App Router + React 19 frontend achieving sub-second load times via Framer Motion hardware-accelerated animations, Radix UI accessible components, and React-Leaflet geospatial visualisation.

F1 Race Prediction & Analytics System

2024

Python · FastAPI · Scikit-learn · Random Forest · Next.js

- Trained a Random Forest model on 75 years of Formula 1 data (25,000+ race results) achieving 76% classification accuracy and MAE of 3.2 positions; engineered circuit-aware features capturing driver form and constructor momentum.

- Deployed full-stack: FastAPI inference backend serving a live, interactive Next.js analytics dashboard with historical trend visualisation and real-time predictions.

AI-Based Deepfake Face Detection System

2024

PyTorch · CNN · Vision Transformers (ViT) · OpenCV · Transfer Learning

- Built a dual-architecture deep learning pipeline — CNN for spatial artefact detection combined with Vision Transformer (ViT) for global attention modelling — to classify real vs. AI-generated faces with improved generalisation via transfer learning and data augmentation.

Sidekick — Campus Social & Peer Matching Platform

2024

Flutter · Firebase · Provider · Google OAuth

- Shipped a cross-platform (Android, iOS, Web) campus application featuring real-time chat, a social feed, and a time-slot-based peer matching system with live queue tracking, secured via Google OAuth and Firebase security rules.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, SQL

Frameworks & Libraries: React 19, Next.js 15, FastAPI, Flask, Flutter, Tailwind CSS 4.0, Framer Motion, Radix UI, Recharts, React-Leaflet

Machine Learning & Data: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Feature Engineering, Model Optimisation

Databases & Infra: MySQL, MongoDB, Redis, Pydantic, Uvicorn, Axios, Git, Postman, Vercel, Render, Azure OpenAI, REST APIs

EDUCATION

PSG College of Technology, Coimbatore

2023 – 2027

B.E. Computer Science & Engineering · CGPA: 8.09 / 10 (up to 5th semester)

Relevant coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Operating Systems, Computer Networks

CERTIFICATIONS & AWARDS

Big Data Analytics & Deep Learning — National University of Singapore · **Awarded with High Commendation**

Algorithmic Toolbox — UC San Diego / Coursera

LEADERSHIP & EXTRACURRICULARS

Head of Design — The Eye Club (Cybersecurity Society), PSG College of Technology

Operations Head, TEDx | Under-Secretary of Creators — Global Leaders Forum (GLF Club)